

SKILLS

- **AI / GenAI:** LLMs, RAG, LangChain, Agentic Workflows, LLM Orchestration, Embeddings, Vector DBs (Pinecone, pgvector), Prompt Engineering, Model Evaluation, HuggingFace
- **Cloud & Infrastructure:** GCP Vertex AI, AWS (Bedrock, SageMaker, Lambda, EKS, S3), Kubernetes, Terraform, Docker, CI/CD, GKE
- **Languages & Applications:** Python, TypeScript, JavaScript, Go, SQL | React, Node.js, REST APIs, GraphQL
- **Architecture & Governance:** Enterprise Architecture, Solution Design, Responsible AI, AI Governance, IAM, Zero-Trust, Sovereign AI, GDPR
- **Data & Observability:** Kafka, Airflow, Elasticsearch, Redis, OpenTelemetry, ELK Stack, Distributed Tracing, Prometheus, SLOs

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect Nov 2022 – Present
 - **Conversational AI and RAG – ASB Assist (Under Development):** Building a natural-language architecture generation capability that converts business requirements into compliant diagrams, injects security patterns including OAuth, Zero-Trust, MCP and agent gateways, and recommends enterprise services through an LLM and retrieval-augmented generation pipeline.
 - **Enterprise Solution Architecture Platform:** Designed and led 0-to-1 production delivery of a patent-pending Node.js/TypeScript and React platform; wrote 99.6% of production code and 100% of core logic, mapping 1,000+ architecture patterns to real-time component selections in air-gapped, security-constrained environments.
 - **Responsible AI and Multi-Cloud Governance:** Defined and operationalised an enterprise AI governance framework spanning model risk, IAM, audit trails, OSFI, GDPR and Sovereign AI constraints across AWS, GCP and OCP; adopted organisation-wide.
 - **Technical Leadership and Stakeholder Alignment:** Led 5 engineers through architecture, development, testing and delivery using code reviews, ADRs and weekly sprints; aligned product, risk and architecture stakeholders and produced reusable schema contracts and onboarding guidance.
 - **Enterprise Observability Foundation:** Architected a Kafka and OpenTelemetry platform ingesting 150 TB/day across 10,000+ systems; standardised telemetry adopted by 20+ engineering teams, enabling AIOps and ML operations use cases without vendor-specific integrations.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Applied AI and Cloud Delivery:** Integrated transformer-based document classification for medical-record parsing, including AWS EKS deployment, inference APIs and production monitoring; reduced manual review overhead by 60%.
 - **0-to-1 Product Engineering:** Built the React/TypeScript frontend, Node.js/Express API, blockchain verification layer and HIPAA-aligned AWS deployment; team scaled to 20+ engineers before acquisition.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
IT Developer / Software Developer Co-op Apr 2022 – Nov 2022
 - **Platform Modernisation:** Led migration of a federal tax-platform UI from Apache Struts to React/TypeScript and contributed to an API layer serving millions of Canadians while enforcing WCAG accessibility standards.

PROJECTS

- **Flight Price Optimization Agent:** Built a reinforcement-learning agent using DQN/DDQN and Python for dynamic pricing; modelled demand curves and achieved an 18% improvement over baseline pricing strategies in simulation.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** India
BE Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020